AI Assisted Coding

Assignment –14.1

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Task 1 – Portfolio Website Design

You are building a personal portfolio website to showcase your work.

Requirements:

• Create sections for About Me, Projects, and Contact.

• Use AI to:

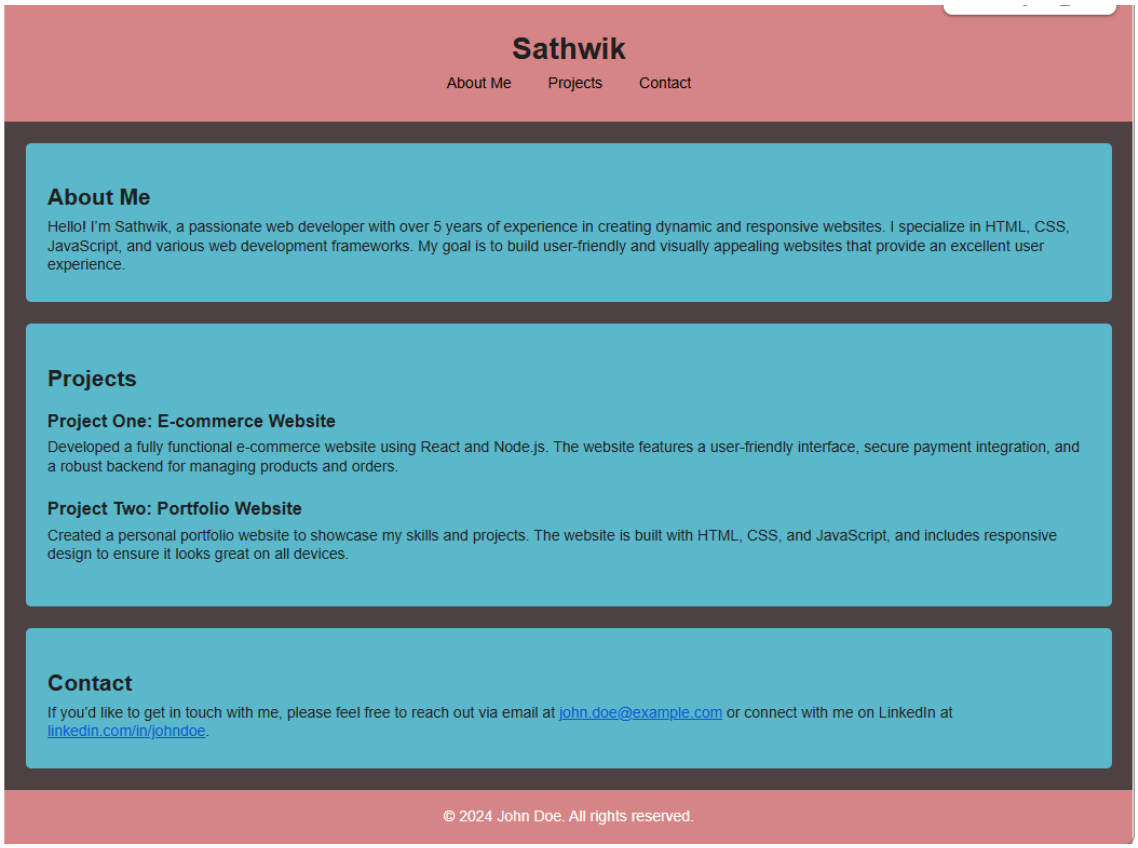
o Suggest color palettes and typography.

o Create a responsive layout with Grid/Flexbox.

o Add smooth scrolling navigation.

```html
<!-- Generate me a professional portfolio using web development technologies -->
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Sathwik - Web Developer Portfolio</title>
    <link rel="stylesheet" href="styles.css">
</head>

<style>
    /* styles.css */
    body {
        font-family: Arial, sans-serif;
        margin: 0;
        padding: 0;
        background-color: #4d4242;
    }

    header {
        background-color: #d58585;
        color: #fff;
        padding: 20px;
        text-align: center;
    }

    nav ul {
        list-style: none;
        padding: 0;
    }

    nav ul li {
        display: inline;
        margin: 0 15px;
    }

    nav ul li a {
        color: #000000;
        text-decoration: none;
    }

    section {
        padding: 20px;
        margin: 20px;
        background-color: #5bb8cb;
        border-radius: 5px;
    }

    .project {
        margin-bottom: 20px;
    }

    footer {
        background-color: #d58585;
        color: #fff;
        text-align: center;
        padding: 10px;
    }
</style>
```

```html
<body>
    <header>
        <h1>Sathwik</h1>
        <nav>
            <ul>
                <li><a href="#about">About Me</a></li>
                <li><a href="#projects">Projects</a></li>
                <li><a href="#contact">Contact</a></li>
            </ul>
        </nav>
    </header>

    <section id="about">
        <h2>About Me</h2>
        <p>Hello! I'm Sathwik, a passionate web developer with over 5 years of experience in creating dynamic and responsive websites. I specialize in HTML, CSS, JavaScript, and various web development frameworks. My goal is to bu
    </section>

    <section id="projects">
        <h2>Projects</h2>

        <div class="project">
            <h3>Project One: E-commerce Website</h3>
            <p>Developed a fully functional e-commerce website using React and Node.js.
                The website features a user-friendly interface, secure payment integration,
                and a robust backend for managing products and orders.</p>
        </div>

        <div class="project">
            <h3>Project Two: Portfolio Website</h3>
            <p>Created a personal portfolio website to showcase my skills and projects.
                The website is built with HTML, CSS, and JavaScript,
                and includes responsive design to ensure it looks great on all devices.</p>
        </div>
    </section>

    <section id="contact">
        <h2>Contact</h2>
        <p>If you'd like to get in touch with me, please feel free to reach out via email at
            <a href="mailto:Sathwik@example.com">john.doe@example.com</a> or connect with me on LinkedIn at
            <a href="https://www.linkedin.com/in/Sathwik" target="_blank">linkedin.com/in/johndoe</a>.
        </p>
    </section>

    <footer>
        <p>&copy; 2024 John Doe. All rights reserved.</p>
    </footer>
</body>
</html>
```



Task 2 – Online Store Product Page

Design a product display page for an online store.

Requirements:

• Display product image, title, price, and "Add to Cart" button.

• Use AI to:

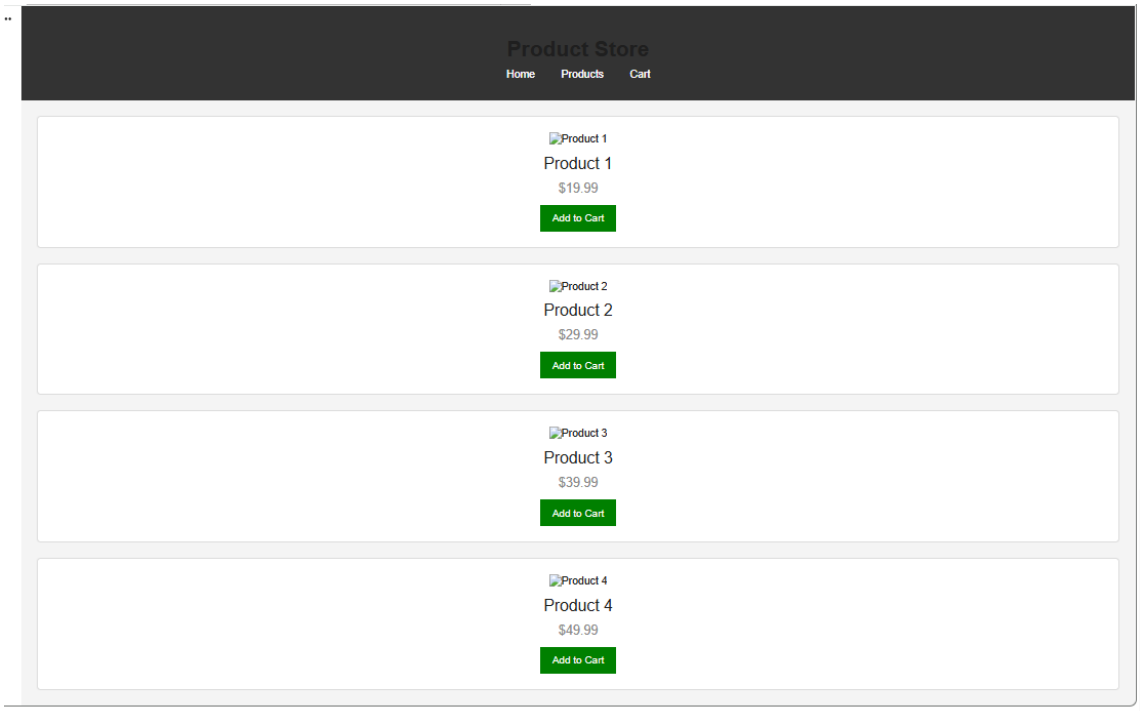
o Style with BEM methodology. o Make

layout responsive. o Add hover e ects and

"Add to Cart" alert

```html
<!-- Create a responsive online store product page using HTML, CSS, and JavaScript that displays a product image, title, price, and an "Add to Cart" button. Use BEM methodology for CSS, add hover effects, responsive layout, and sh

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Product Page</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 0;
            padding: 0;
            background-color: #f4f4f4;
        }

        header {
            background-color: #333;
            color: #fff;
            padding: 20px;
            text-align: center;
        }

        nav ul {
            list-style: none;
            padding: 0;
        }

        nav ul li {
            display: inline;
            margin: 0 15px;
        }

        nav ul li a {
            color: #fff;
            text-decoration: none;
        }

        .product {
            background-color: #fff;
            border: 1px solid #ddd;
            border-radius: 5px;
            padding: 20px;
            margin: 20px;
            text-align: center;
        }

        .product img {
            max-width: 100%;
            height: auto;
        }

        .product-title {
            font-size: 1.5em;
            margin: 10px 0;
        }

        .product-price {
            color: #888;
```

```
            color: #fff;
            border: none;
            padding: 10px 15px;
            cursor: pointer;
        }

        .add-to-cart:hover {
            background-color: darkgreen;
        }

        @media (max-width: 600px) {
            .product-price {
                font-size: 1em;
            }
        }
    </style>
</head>

<body>

<header>
    <h1>Product Store</h1>
    <nav>
        <ul>
            <li><a href="#home">Home</a></li>
            <li><a href="#products">Products</a></li>
            <li><a href="#cart">Cart</a></li>
        </ul>
    </nav>
</header>

<section id="products">

    <div class="product">
        <img src="product1.jpg" alt="Product 1">
        <div class="product-title">Product 1</div>
        <div class="product-price">$19.99</div>
        <button class="add-to-cart" onclick="addToCart('Product 1')">Add to Cart</button>
    </div>

    <div class="product">
        <img src="product2.jpg" alt="Product 2">
        <div class="product-title">Product 2</div>
        <div class="product-price">$29.99</div>
        <button class="add-to-cart" onclick="addToCart('Product 2')">Add to Cart</button>
    </div>

    <div class="product">
        <img src="product3.jpg" alt="Product 3">
        <div class="product-title">Product 3</div>
        <div class="product-price">$39.99</div>
        <button class="add-to-cart" onclick="addToCart('Product 3')">Add to Cart</button>
    </div>

    <div class="product">
        <img src="product4.jpg" alt="Product 4">
        <div class="product-title">Product 4</div>
        <div class="product-price">$49.99</div>
        <button class="add-to-cart" onclick="addToCart('Product 4')">Add to Cart</button>
    </div>

</section>

<script>
    function addToCart(productName) {
        alert(productName + " has been added to your cart!");
    }
</script>

</body>
</html>
```



Task 3 – Event Registration Form

Build an event registration form for a conference.

Requirements:

• Collect name, email, phone number, and session selection.

• Use AI to:

o Add form validation with JavaScript. o Make the form

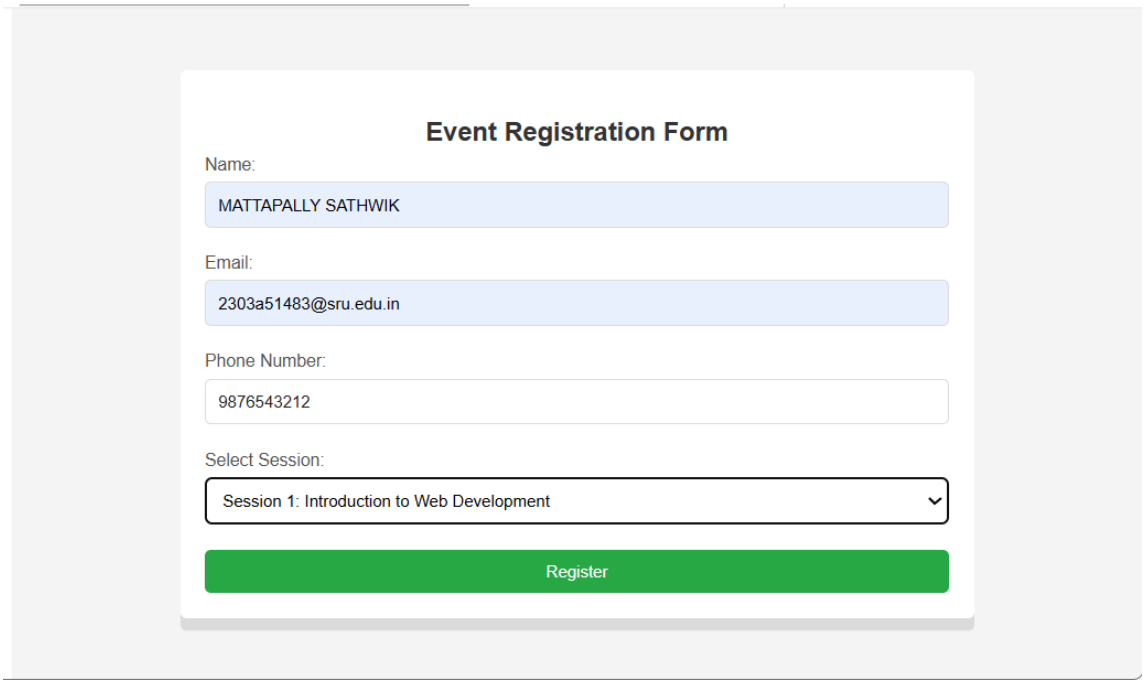
accessible with labels and ARIA.

o Style with a professional look

```html
<!-- Create an Event Registration Form using HTML, CSS, and JavaScript that collects name, email, phone number, and session selection. Add JavaScript form validation, ensure accessibility using labels and ARIA attributes, and style the form with a clean professional design. -->

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Event Registration Form</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f4f4f4;
            margin: 0;
            padding: 0;
        }

        .container {
            max-width: 600px;
            margin: 50px auto;
            background-color: #fff;
            padding: 20px;
            border-radius: 5px;
            box-shadow: 0 10px rgba(0, 0, 0, 0.1);
        }

        h2 {
            text-align: center;
            color: #333;
        }

        form {
            display: flex;
            flex-direction: column;
        }

        label {
            margin-bottom: 5px;
            color: #555;
        }

        input, select {
            padding: 10px;
            margin-bottom: 20px;
            border: 1px solid #ddd;
            border-radius: 5px;
        }

        button {
            padding: 10px;
            background-color: #28a745;
            color: #fff;
            border: none;
            border-radius: 5px;
            cursor: pointer;
        }

        button:hover {
            background-color: #218838;
        }
    </style>
</head>

<body>
    <div class="container">
        <h2>Event Registration Form</h2>

        <form id="registrationForm" aria-label="Event Registration Form">

            <label for="name">Name:</label>
            <input type="text" id="name" name="name" aria-required="true" required>

            <label for="email">Email:</label>
            <input type="email" id="email" name="email" aria-required="true" required>
```

Open Website | Ln 300, Col 8 | Spaces: 4 | UTF-8 | CRLF | {} Python | Python 3.13 | Go Live

```html
Click to add a breakpoint
            <input type="email" id="email" name="email" aria-required="true" required>

            <label for="phone">Phone Number:</label>
            <input type="tel" id="phone" name="phone" aria-required="true" required pattern="[0-9]{10}">

            <label for="session">Select Session:</label>
            <select id="session" name="session" aria-required="true" required>
                <option value="">Please choose an option</option>
                <option value="session1">Session 1: Introduction to Web Development</option>
                <option value="session2">Session 2: Advanced JavaScript</option>
                <option value="session3">Session 3: CSS for Beginners</option>
            </select>

            <button type="submit">Register</button>
        </form>
    </div>

    <script>
        document.getElementById("registrationForm").addEventListener("submit", function(event) {
            event.preventDefault();

            const name = document.getElementById("name").value.trim();
            const email = document.getElementById("email").value.trim();
            const phone = document.getElementById("phone").value.trim();
            const session = document.getElementById("session").value;

            if (!name || !email || !phone || !session) {
                alert("Please fill in all fields.");
                return;
            }

            if (!/^[0-9]{10}$/.test(phone)) {
                alert("Please enter a valid 10-digit phone number.");
                return;
            }

            alert(`Thank you for registering, ${name}! You have selected ${session}.`);
        });
    </script>
</body>
</html>
```



Task 4 – Resume Website Build a

one-page online resume.

Requirements:

• Sections: Profile Summary, Skills, Education, Experience.

• Use AI to:

o Propose professional typography styles.

o Align sections using CSS Grid. o Add smooth scroll

navigation between sections.

```html
<!-- Create a one-page resume website using HTML, CSS, and JavaScript with sections for Profile Summary, Skills, Education, and Experience. Use professional typography styles, align sections using CSS Grid, and implement smooth scrolling navigation between sections. -->

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Sathwik's Resume</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 0;
            padding: 0;
            background-color: #f4f4f4;
        }

        header {
            background-color: #333;
            color: #fff;
            padding: 20px;
            text-align: center;
        }

        nav ul {
            list-style: none;
            padding: 0;
        }

        nav ul li {
            display: inline;
            margin: 0 15px;
        }

        nav ul li a {
            color: #fff;
            text-decoration: none;
        }

        section {
            padding: 20px;
            margin: 20px;
            background-color: #fff;
            border-radius: 5px;
        }

        h2 {
            color: #333;
        }
    </style>
</head>

<body>
    <header>
        <h1>Sathwik's Resume</h1>
        <nav>
            <ul>
                <li><a href="#profile">Profile Summary</a></li>
                <li><a href="#skills">Skills</a></li>
                <li><a href="#education">Education</a></li>
                <li><a href="#experience">Experience</a></li>
            </ul>
        </nav>
    </header>

    <section id="profile">
        <h2>Profile Summary</h2>
        <p>Experienced Web Developer with a strong background in HTML, CSS, and JavaScript. Passionate about creating responsive and user-friendly websites.</p>
    </section>

    <section id="skills">
        <h2>Skills</h2>
        <ul>
```

O

```html
            <ul>
                <li>HTML5 & CSS3</li>
                <li>JavaScript & jQuery</li>
                <li>Responsive Design</li>
                <li>Version Control (Git)</li>
            </ul>
        </section>

        <section id="education">
            <h2>Education</h2>
            <p>Bachelor of Science in Computer Science, XYZ University, 2020</p>
        </section>

        <section id="experience">
            <h2>Experience</h2>
            <p>Web Developer at ABC Company, 2020-Present</p>
        </section>

    </body>
    </html>
```

Sathwik's Resume

Profile Summary Skills Education Experience

Profile Summary

Experienced Web Developer with a strong background in HTML, CSS, and JavaScript. Passionate about creating responsive and user-friendly websites.

Skills

- HTML5 & CSS3
- JavaScript & jQuery
- Responsive Design
- Version Control (Git)

Education

Bachelor of Science in Computer Science, XYZ University, 2020

Experience

Web Developer at ABC Company, 2020-Present

Task 5 – Restaurant Menu Page

Design a digital menu page for a restaurant.

Requirements:

• Display menu items with name, description, and price.

• Use AI to:

o Suggest a food-friendly color palette. o

Design a responsive card-based layout. o

Add hover e ects on menu items.

Lab14 > task5.html > html > body > div.menu > div.menu-item

```html
<!--Create a restaurant digital menu page using HTML and CSS that
 displays menu items with name, description, and price in a responsive
 card-based layout. Use a food-friendly color palette and add
 hover effects on menu items for better user interaction. -->
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Restaurant Menu</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #fff8f0;
            margin: 0;
            padding: 0;
        }
        header {
            background-color: #ff6347;
            color: #fff;
            padding: 20px;
            text-align: center;
        }
        .menu {
            display: flex;
            flex-wrap: wrap;
            justify-content: center;
            padding: 20px;
        }
        .menu-item {
            background-color: #fff;
            border: 1px solid #ddd;
            border-radius: 5px;
            width: 300px;
            margin: 15px;
            padding: 20px;
            text-align: center;
            transition: transform 0.3s ease, box-shadow 0.3s ease;
        }
        .menu-item:hover {
            transform: translateY(-10px);
            box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
        }
        .menu-item h3 {
            margin-top: 0;
        }
        .menu-item p {
            color: #555;
```

Lab14 > task5.html > html > body > div.menu > div.menu-item > p

```html
<html lang="en">
<head>
    <style>
        .menu-item h3 {
        }
        .menu-item p {
            color: #555;
        }
        .menu-item .price {
            color: #ff6347;
            font-size: 1.2em;
            margin-top: 10px;
        }
    </style>
</head>
<body>
    <header>
        <h1>Delicious Bites</h1>
    </header>
    <div class="menu">
        <div class="menu-item">
            <h3>Grilled Salmon</h3>
            <p>Fresh Atlantic salmon grilled to perfection, served with seasonal
                vegetables.</p>
            <div class="price">$18.99</div>
        </div>
        <div class="menu-item">
            <h3>Caesar Salad</h3>
            <p>Crisp romaine lettuce with homemade Caesar dressing and croutons.</p>
            <div class="price">$12.99</div>
        </div>
        <div class="menu-item">
            <h3>Margherita Pizza</h3>
            <p>Classic pizza with fresh tomatoes, mozzarella, and basil.</p>
            <div class="price">$14.99</div>
        </div>
        <div class="menu-item">
            <h3>Chocolate Lava Cake</h3>
            <p>Decadent chocolate cake with a molten chocolate center.</p>
            <div class="price">$8.99</div>
        </div>

    </div>
</body>
</html>
```

Delicious Bites

Grilled Salmon

Fresh Atlantic salmon grilled to perfection, served with seasonal vegetables.

$18.99

Caesar Salad

Crisp romaine lettuce with homemade Caesar dressing and croutons.

$12.99

Margherita Pizza

Classic pizza with fresh tomatoes, mozzarella, and basil.

$14.99

Chocolate Lava Cake

Decadent chocolate cake with a molten chocolate center.

$8.99